

AIGR Deep Research Report — *S. pombe*

cps1 / SPAC24C9.08 (UniProt O13968)

Focus type: function_assignment **Hypothesis slug:** additional-lipid-amide-and-extracellular-capacities **Gene:** cps1 (Carboxypeptidase S, CBPS_SCHPO), *Schizosaccharomyces pombe* 972h- (NCBITaxon:284812)

Executive Judgment

Verdict: Over-annotated / weakly supported for the "additional" functions; the core vacuolar carboxypeptidase function is well supported.

Evaluated claim-by-claim, the seed's proposed *additional* capacities for O13968 are **not** backed by direct evidence:

Claim	Verdict
Nonpeptide N-acyl-amino-acid hydrolase/synthase (PM20D1-type MF)	Over-annotated / refuted as direct evidence — electronic-only (IEA:UniProtKB-ARBA), driven by the PANTHER family <i>name</i> PM20D1; no fungal assay.
Lipid metabolic process (GO:0006629)	Over-annotated — IEA:UniProtKB-ARBA family transfer only.
Extracellular localization (GO:0005576)	Refuted / over-annotated — IEA:UniProtKB-ARBA; contradicted by HDA vacuolar localization and type II luminal topology.
Adaptive thermogenesis (GO:1990845)	Refuted — not annotated at all; no mechanistic basis in a unicellular fungus; PM20D1 thermogenesis is brown/beige-adipocyte-specific.
Vacuolar metallocarboxypeptidase (core)	Supported — HDA/TAS/IBA/ISO evidence, M20A family, defined active site and Gly-penultimate specificity.

The seed is correct on several methodological points: (a) the target is *not* bgs1 glucan synthase; (b) the PANTHER family label "PM20D1" is not proof of transferred specificity; (c) GFP-/Ub-GFP-constructs report engineered MVB cargo behaviour, not native secretion; (d) absence of a fungal biochemical assay is not positive proof of loss of any activity. However, the *practical curation consequence* is that the additional PM20D1-derived MF/BP/CC terms rest solely on rule-based electronic inference and should not be treated as gene-product functions without experimental support. The primary, experimentally-anchored function is a **luminal vacuolar-membrane metallocarboxypeptidase (peptidase M20A)**.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Con
UniProt O13968 (CBPS_SCHPO), DB record	Review/ database	Refutes "extra" claims; supports core	Function & localization	Curated as "Carboxypeptidase S / Vacuolar carboxypeptidase cps1"; peptidase M20A ; catalytic activity = release of C-terminal residue when Gly is penultimate; Vacuole membrane	S. po
UniProt O13968 GO evidence codes (computed check)	Database/ computational	Qualifies (provenance)	Which GO terms are experimental	Vacuole/carboxypeptidase terms = HDA/TAS/IBA/ISO; GO:0005576 extracellular, GO:0006629 lipid, GO:0016810 C-N non-peptide hydrolase = IEA:UniProtKB- ARBA; no GO:1990845	S. po
UniProt O13968 topology (computed)	Structural/ computational	Refutes secretion/ thermogenesis	Localization/ mechanism	Type II membrane protein: cyto 1–37, TM 38–58, lumenal 59–596 ; catalytic Zn-peptidase domain faces vacuole lumen	S. po
Pairwise NW identity (computed)	Structural/ evolutionary	Qualifies (paralog distance)	"Same family ⇒ same function?"	O13968 vs ScCPS1 = 36.7% ; O13968 vs human PM20D1 = 27.9% ; ScCPS1 vs PM20D1 = 27.5%	cross spec
 27374330 (Long et al., 2016, <i>Cell</i>)	Direct assay (mammal)	Competing / non- transferable	Origin of PM20D1 activity	PM20D1 is a secreted enzyme enriched in UCP1+ adipocytes; bidirectional N-acyl-amino-acid synthase/hydrolase; products uncouple mitochondria	Mou hum adip
	Localization/ mutant (yeast)	Qualifies		GFP-Cps1 is the canonical ubiquitin-dependent MVB	S. cere

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Con
P 18508771 (Ren et al., 2008)			GFP-Cps1 / Ub constructs	cargo delivered to the vacuole lumen; missorted when Ub sorting is disrupted	
P 31341193 (Yanguas et al., 2019)	Localization/ genetic (fission yeast)	Supports core	Native trafficking	<i>S. pombe</i> Cps1 follows the CPY pathway TGN→PVE→ vacuole via GGA/Ent3 adaptors	<i>S. po</i>
InterPro member DBs for O13968 (computed)	Structural/ evolutionary; database	Refutes PM20D1 assignment	Family/ specificity	CDD cd05674 carboxypeptidase yscS , PIRSF037217 Carboxypeptidase S , IPR017141→GO:0004181; PANTHER PTHR45962 (PM20D1) carries no GO	cross spec signa
Catalytic-residue mapping (computed, NW)	Structural/ evolutionary	Qualifies	"Same family ⇒ same specificity?"	M20 Zn/catalytic core (H197,D199,D232,E266,E267,E296) 100% conserved in all three; substrate residue Sp H565 = ScCPS1 H547 but = PM20D1 N467 (divergent)	O139 P276 Q6G

Residue-level & domain-database analysis (Iteration 2, computed)

- **Independent domain databases assign carboxypeptidase S, not PM20D1.** For O13968, InterPro member DBs give: CDD **cd05674 "M20 Peptidase, carboxypeptidase yscS"**; PIRSF037217 **"Carboxypeptidase S"**; InterPro IPR017141 **"Peptidase M20, carboxypeptidase S"** → GO:0004181 **metallocarboxypeptidase activity**. The PANTHER PTHR45962 entry (name "N-fatty-acyl-amino acid synthase/hydrolase PM20D1") carries **no GO terms** — the PM20D1 label is a family name without functional annotation.
- **M20 catalytic core is 100% conserved** across O13968 / ScCPS1 / human PM20D1: His197, Asp199, Asp232, Glu266 (proton acceptor), Glu267, Glu296 all align 1:1 (PM20D1 His125/Asp127/Asp157/Glu191/Glu192/Glu218). ⇒ All three are genuine binuclear-zinc amide-bond

metallohydrolases; O13968 mechanistically **retains** amide/peptide hydrolase capability (argues against "loss of peptide hydrolysis").

- **A substrate-binding residue tracks the fungal CPS clade, not PM20D1:** O13968 **His565** = **ScCPS1 His547 (match)** but = **PM20D1 Asn467 (divergent)**. Consistent with conserved carboxypeptidase-S substrate recognition in the fungal enzymes and altered specificity in PM20D1.

Net: shared catalytic chemistry does not equal shared substrate specificity. The specificity-informative, GO-bearing signatures unanimously call O13968 a carboxypeptidase S.

GO Curation Implications (leads — require curator verification)

Retain (core, experimentally/phylogenetically supported): - **MF** GO:0004181 metallocarboxypeptidase activity (ISO) and GO:0004180 carboxypeptidase activity (IBA) — retain; these are the direct molecular function. - **CC** GO:0000324 fungal-type vacuole (HDA) / GO:0000328 vacuole lumen (IBA) / GO:0005774 vacuolar membrane — retain (type II membrane, luminal catalytic domain). - **BP** GO:0007039 protein catabolic process in the vacuole (TAS) / GO:0006520 amino acid metabolic process — retain.

Treat as non-core / candidate for removal or "do not accept" (IEA:UniProtKB-ARBA only, family-name driven): - **GO:0005576 extracellular region** — flag as over-annotation; conflicts with HDA vacuole localization and luminal type II topology. Lead: do **not** promote to a curated CC; consider NOT-qualifier or removal. - **GO:0006629 lipid metabolic process** — flag as over-annotation; no fungal lipid-catalysis evidence. - **GO:0016810 hydrolase activity acting on C–N (but not peptide) bonds** — flag; this is the N-acyl-amino-acid-hydrolase-flavored MF and is electronic-only.

Do not add: - **GO:1990845 adaptive thermogenesis** — no evidence, mechanistically inapplicable to a unicellular fungus; PM20D1 thermogenesis is adipocyte/mitochondrial-context specific. Recommend against.

Avoid "protein binding" as an outcome — the informative MF here is **metallocarboxypeptidase activity**.

GO Decision Table (computed provenance — see [go_decision_table.csv](#))

GO_ID	Term	Aspect	Current evidence	Curation lead	Flag
GO:0004181	metallocarboxypeptidase activity	MF	ISO	RETAIN (primary MF)	—
GO:0004180	carboxypeptidase activity	MF	IBA	RETAIN	—
GO:0000324	fungus-type vacuole	CC	HDA:PomBase	RETAIN	—
GO:0000328	fungus-type vacuole lumen	CC	IBA	RETAIN	—
GO:0005774	vacuolar membrane	CC	IEA-SubCell	RETAIN	—
GO:0007039	protein catabolic process in vacuole	BP	TAS:PomBase	RETAIN	—
GO:0006520	amino acid metabolic process	BP	IEA-ARBA	RETAIN	—
GO:0005576	extracellular region	CC	IEA:UniProtKB-ARBA	REMOVE / NOT	OVER-ANNOTATION
GO:0006629	lipid metabolic process	BP	IEA:UniProtKB-ARBA	REMOVE / non-core	OVER-ANNOTATION
GO:0016810	hydrolase acting on C–N (non-peptide) bonds	MF	IEA:UniProtKB-ARBA	REMOVE / non-core	OVER-ANNOTATION
GO:1990845	adaptive thermogenesis	BP	(absent)	DO NOT ADD	REFUTED

Native

localization

support:

P 16823372

(Matsuyama et al., 2006) — genome-wide YFP ORFeome localization of ~90% of the *S. pombe* proteome — provides the native (non-engineered) localization dataset the seed references; it is a high-throughput localization resource consistent with the HDA vacuolar assignment (as opposed to the engineered GFP-/Ub-GFP-Cps1 MVB-cargo constructs).

Mechanistic Scope

Direct molecular function tested: zinc-dependent M20A exopeptidase releasing a C-terminal amino acid from peptides with a penultimate glycine (carboxypeptidase S activity), acting in the vacuole lumen for peptide/amino-acid recycling and use of peptides as nitrogen source.

Downstream / non-core: MVB sorting behaviour of GFP-/Ub-GFP-Cps1 fusions is a property of ubiquitin-dependent cargo trafficking machinery, **not** of the enzyme's catalytic activity, and does not indicate secretion. The PM20D1 "uncoupling/thermogenesis" phenotype is a mammalian, whole-organism metabolic outcome of a *secreted* paralog and is not a molecular property attributable to O13968.

Conflicts and Alternatives

- **Paralog / family-name carry-over:** PANTHER PTHR45962 is named for mammalian PM20D1; UniProt's ARBA rules propagate lipid-amide/extracellular terms to all members, including this fungal M20A vacuolar carboxypeptidase. This is the classic paralog-overannotation and frequency-bias pattern.
- **Name collision:** "cps1" also denotes *S. pombe* **bgs1/SPBC19G7.05c** (1,3- β -glucan synthase). The seed correctly excludes it; this report concerns only SPAC24C9.08/O13968.
- **Topology conflict:** A secreted, circulating N-acyl-amino-acid enzyme (PM20D1) vs a type II vacuolar-membrane protein with luminal catalytic domain — the localizations are mutually exclusive.
- **Divergence:** ~28% identity to PM20D1 (twilight zone) vs ~37% to a bona fide fungal carboxypeptidase S argues the substrate specificity should be read from the fungal CPS clade.

Knowledge Gaps

1. **No fungal biochemical assay** of O13968 (or ScCPS1) for N-acyl-amino-acid synthase/hydrolase activity exists. *Checked:* UniProt catalytic annotation + literature; only peptidase

activity is documented. *Matters*: one cannot positively exclude a promiscuous side-activity; but absence of evidence $\neq$ presence of function for curation. *Resolve*: in vitro assay with purified enzyme against N-acyl-amino-acid panel.

2. **Active-site substrate-determinant comparison** with PM20D1 not performed residue-by-residue here. *Matters*: would test whether the fatty-acyl binding pocket exists. *Resolve*: structural/AlphaFold pocket comparison and specificity-residue alignment.
3. **Native secretion**: no evidence of extracellular pool; HDA is vacuolar. *Resolve*: secretome/immunolocalization in *S. pombe*.

Discriminating Tests

- **In vitro activity panel**: purified O13968 assayed for (i) Gly-penultimate carboxypeptidase activity (expected positive) and (ii) N-oleoyl-/N-arachidonoyl-amino-acid hydrolysis/condensation (tests PM20D1-type activity). Bidirectional LC-MS assay as in

P 27374330

- **Structure-guided pocket analysis**: AlphaFold model + superposition on PM20D1/carboxypeptidase-S structures; check for a hydrophobic acyl-binding channel and conserved Zn-coordinating residues (His199/His-cluster, Glu266 proton acceptor already annotated).
- **Localization**: endogenous-tag microscopy and secretome MS to confirm vacuolar (not extracellular) steady-state pool.
- **Specificity-residue alignment** across M20A CPS clade vs M20 PM20D1 clade to identify divergent substrate determinants.

Curation Leads (require curator verification)

1. **Downgrade/flag electronic PM20D1-derived terms** (GO:0005576, GO:0006629, GO:0016810) as IEA-only family carry-over; do not elevate to experimental status. Suggested action: mark as over-annotation / non-core; consider removal or NOT for extracellular.
2. **Do not add GO:1990845 (adaptive thermogenesis)**. No support; mechanistically inapplicable.

3. **Retain and foreground** MF GO:0004181/GO:0004180 (metallocarboxypeptidase), CC vacuole/vacuolar-lumen terms, BP vacuolar protein catabolism.
 4. **Candidate references to verify (with snippets):**
 5. PMID **27374330** — "we identify a secreted enzyme, peptidase M20 domain containing 1 (PM20D1) ... catalyzing both the condensation of fatty acids and amino acids to generate N-acyl amino acids and also the reverse hydrolytic reaction." → establishes activity in a *secreted mammalian adipocyte* context, not transferable by homology.
 6. PMID **31341193** — "Carboxypeptidases Y (Cpy1) and S (Cps1) ... follow the carboxypeptidase Y (CPY) pathway from the trans-Golgi network (TGN) to the prevacuolar endosome (PVE)." → native *S. pombe* Cps1 traffics to the vacuole.
 7. PMID **18508771** — "missorting of the MVB cargo GFP-Cps1." → GFP-Cps1 is engineered vacuolar MVB cargo, not evidence of secretion.
 8. **Suggested question for curators:** Should the *S. pombe* CBPS record's ARBA-derived extracellular/lipid/C–N-hydrolase terms be suppressed pending a fungal biochemical assay?
 9. **Suggested experiment:** bidirectional N-acyl-amino-acid LC-MS assay on purified O13968 to positively test (not assume) PM20D1-type promiscuity.
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Provenance

Computed checks (UniProt REST retrieval of O13968 annotations/topology/GO evidence codes; Needleman-Wunsch pairwise identities O13968/P27614/Q6GTS8) were executed in this session; identities 36.7% (vs ScCPS1) and 27.9% (vs human PM20D1) are direct outputs. Simple match/mismatch scoring was used, so absolute identities are approximate; the relative ordering (closer to fungal CPS than to PM20D1) is the robust conclusion. Literature evidence retrieved via PubMed (PMIDs 27374330, 31341193, 18508771).
